Slope – Deflection Equation:

General:

Slope – deflection equation is one of the methods of analysis of statically indeterminate structures and serves as a basis for other methods (like moment distribution, etc). In this method, the moments at the ends of the member are expressed in terms of the rotation and relative displacements at the ends of the member. The unknown end rotation and displacements are determined by considering equilibrium condition of joints, at which two or members of the structure join. The slope deflection method is more easily carried out and suitable for structures with few numbers of unknown displacements and large numbers of redundant supports.

The" ***bending moment sign convention***" used in identifying the moments as positive or negative for analysis of beams is not suitable in the analysis of generally frames using **SD** or other advanced method. Accordingly the following **statical sign** convention, simple to understand and avoid confusions, is established based on joint and member end rotation.

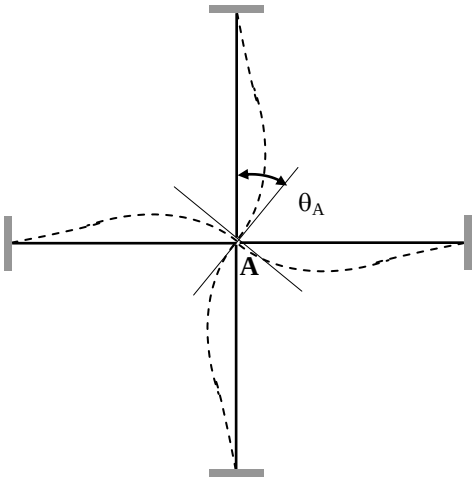
Statical Sign Convention:

(a) The rotation θ of a joint or member is positive if the tangent turns in clock wise direction, and the opposite sense is considered as negative

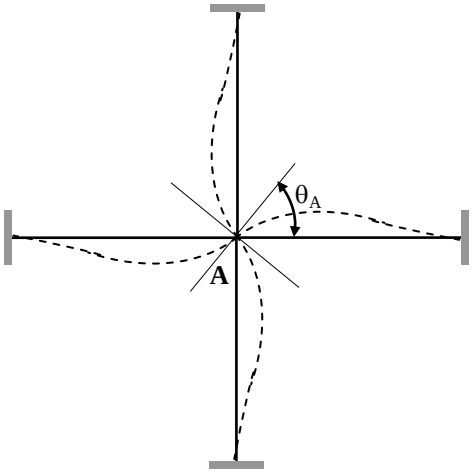
(b) An end moment M is considered positive if it tends to rotate the member counter clockwise or the joint clockwise and the opposite sense is considered as negative.

The positive and negative sense of rotation and corresponding moments is shown in the following diagrams.

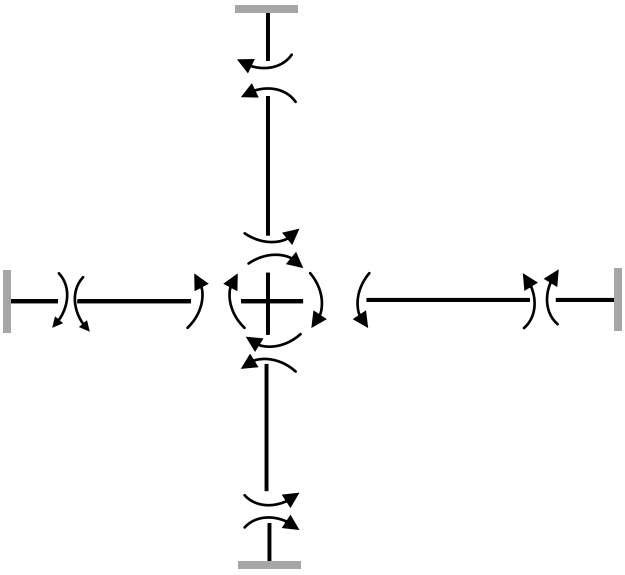
Statical Sign Convention



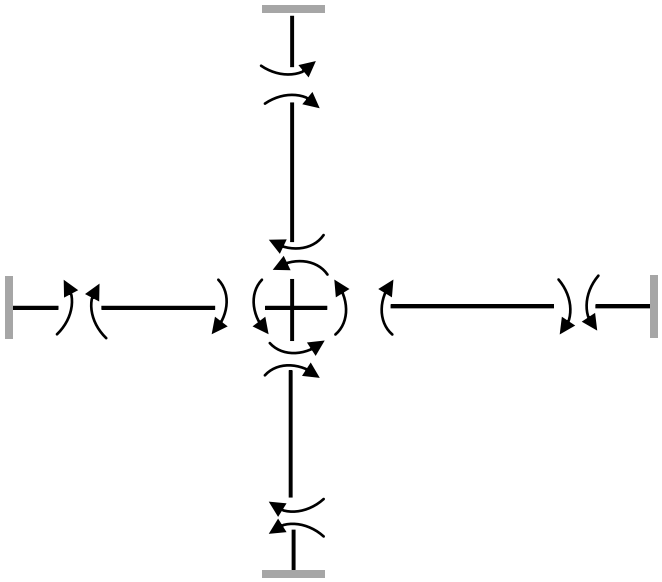
Positive Rotation



Negative Rotation



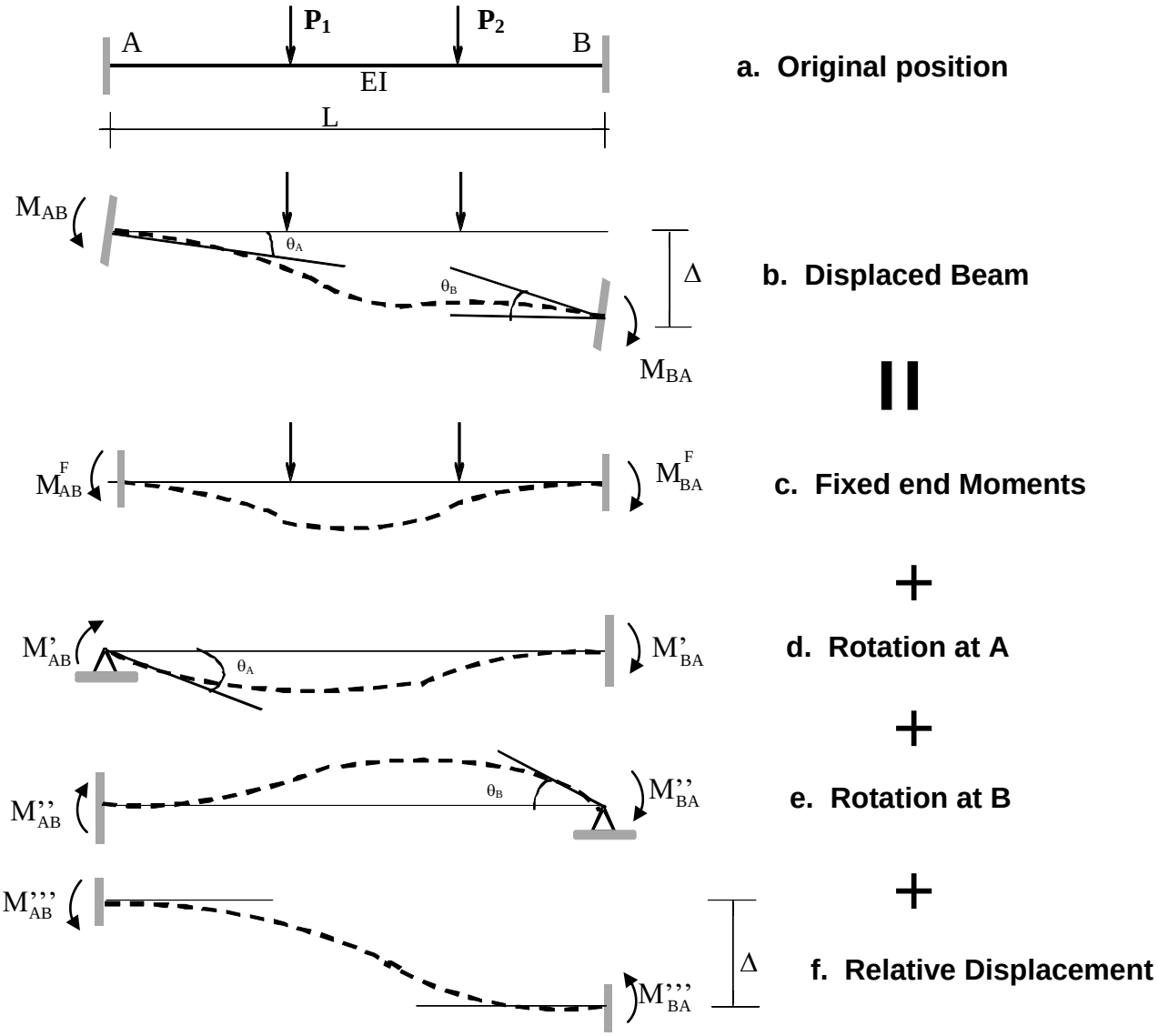
Positive Bending Moment



Negative Bending Moment

Derivation of Slope – Deflection Equations.

Consider a prismatic and restrained beam AB as shown is fig. (a) below with constant flexural stiffness EI. Assume that the beam will undergo end rotations θ_A and θ_B and relative displacement Δ due to the applied loads as shown in fig. (b)



The beam in fig.(b) is equivalent to the sum of the states of loading shown in fig. c to f above . These components are:

i) Fixed end moments $\Leftrightarrow$ fig (c)

ii) End moments due to rotation at the supports, θ_A and θ_B $\Leftrightarrow$ fig.(d) & fig (e)

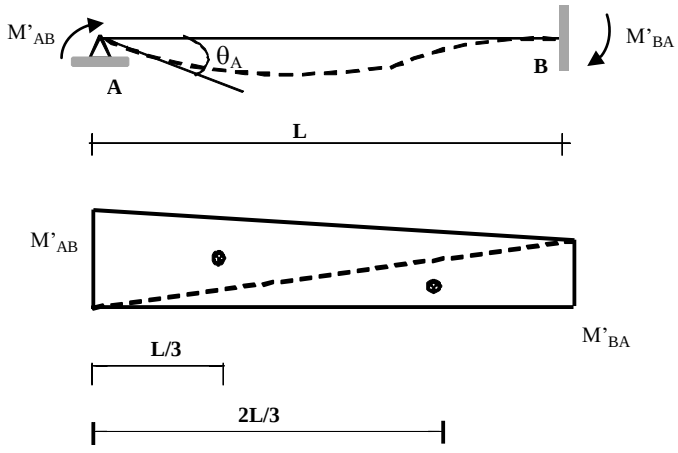
iii) End moments due relative displacement Δ of the ends $\Leftrightarrow$ fig. (f)

The fixed end moments for various loadings can be determined by using either the three-moment equation or consistent deformation. The end moments due to rotations and relative displacement of the ends can be determined by using the moment area method. A positive sense of rotation and relative displacement are considered and the actual sense of moments at the ends corresponding to these rotation and relative displacement are indicated. The magnitude of corresponding end moments are computed using moment area method as per bending moment sign convention and the actual sense of the end moments with respect to the statical sign convention can be identified later on.

Fixed-End Moments:

When the ends of a beam are completely fixed against rotation and translation under the action of transverse loads, the beam is called a ***fixed-end beam***. The corresponding end moments at the supports are called fixed end moments. The fixed end moments for most commonly encountered cases of loading can be determined using suitable methods(like moment area method or the three moment equation ,etc) and are normally given in tabular form parametrically in different literatures.

End Moment Due to Rotation at A, θ_A



From the second t theorem of moment area method, and adopting the bending moment sign convention,

$$t_{AB} = \Delta = \frac{M'_{AB}L}{2EI}(\frac{L}{3}) + \frac{M'_{BA}L}{2EI}(\frac{2L}{3}) = 0$$

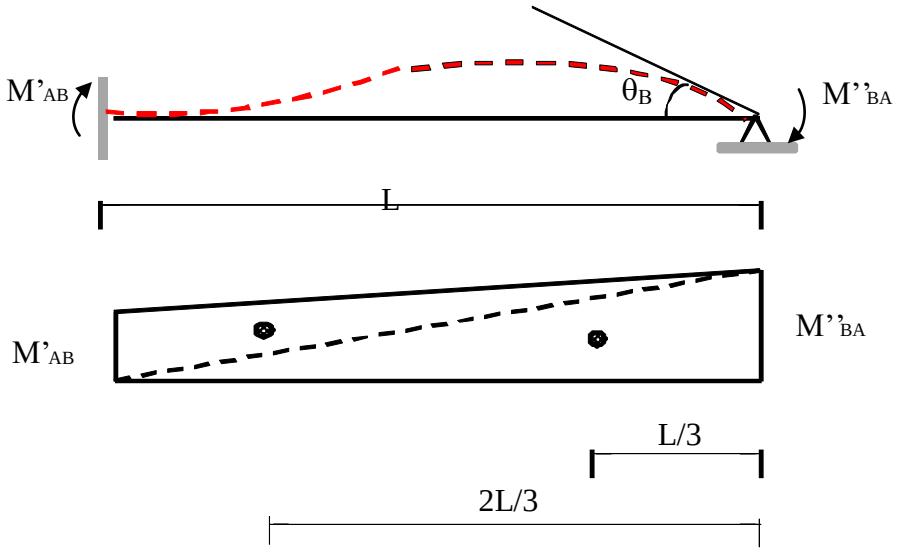
Using the first moment-area theorem,

$$\theta_A - \theta_B = \theta_A = \frac{(M'_{AB} + M'_{BA})L}{2EI} = \frac{(M'_{AB} - \frac{M'_{AB}}{2})L}{2EI} \qquad \rightarrow \theta_A = \frac{M'_{AB}}{4EI}L$$

Hence

$$M'_{AB} = \frac{4EI\theta_A}{L} \qquad and \qquad M'_{BA} = -\frac{2EI\theta_A}{L}$$

End Moment Due to Rotation at B, θ_B



In a manner similar to the above,

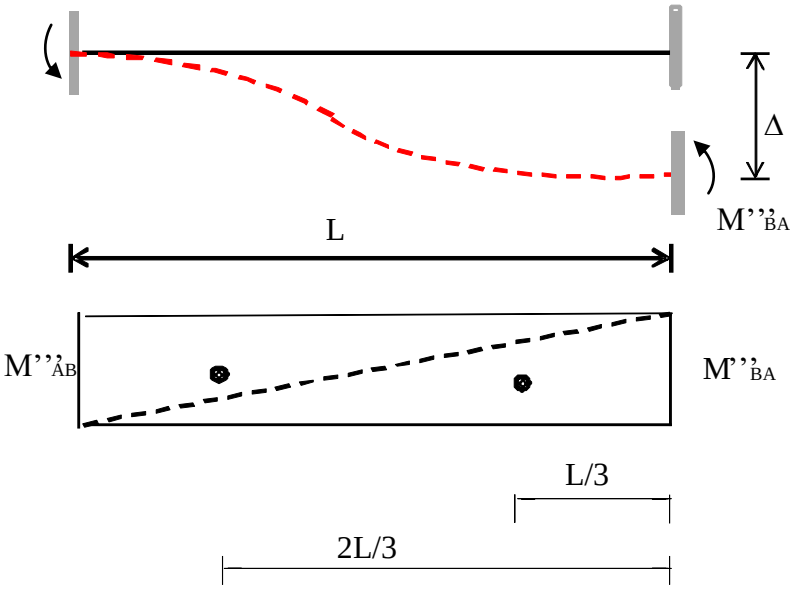
$$\theta_B = \frac{(M''_{AB} + M''_{BA})L}{2EI} \qquad \text{and} \qquad \Delta_B = \frac{M''_{BA}L}{2EI}(\frac{L}{3}) + \frac{M''_{AB}L}{2EI}(\frac{2L}{3}) = 0$$

From which

$$M''_{AB} = -\frac{1}{2}M''_{BA} \qquad \text{and} \qquad \theta_B = \frac{M''_{BA}L}{4EI}$$

Hence $M''_{BA} = \frac{4EI\theta_B}{L}$ and $M''_{AB} = -\frac{2EI\theta_B}{L}$

End Moments due to Relative Displacement Δ



From the first theorem of moment-area method,

$$\theta_A - \theta_B = \frac{(M'''_{AB} + M'''_{BA})L}{2EI} = 0 \qquad thus \qquad M'''_{AB} = -M'''_{BA}$$

Using the second theorem of the moment area method,

$$t_{BA} = \Delta = \frac{M'''_{BA}L}{2EI}(\frac{L}{3}) + \frac{M'''_{AB}L}{2EI}(\frac{2L}{3})$$

Hence

$$M'''_{AB} = -\frac{6EI}{L^2}\Delta \qquad and \qquad M'''_{BA} = \frac{6EI}{L^2}\Delta$$

After determining the magnitude, it is important to note that the moments shown above are with exact sense corresponding to positive sense of rotation θ and relative displacement Δ. Positive and negative sense of moments as per the statical sign convention is then identified to be as follows. Introducing geometric stiffness of the member **K = I/L**

$$M'_{AB} = -\frac{4EI}{L}\theta_A = -4EK\,\theta_A \quad ; \quad M'_{BA} = -\frac{2EI}{L}\theta_A = -2EK\,\theta_A$$

$$M''_{AB} = -\frac{2EI}{L}\theta_B = -2EK\,\theta_B \quad ; \quad M''_{BA} = -\frac{4EI}{L}\theta_B = -4EK\,\theta_B$$

$$M'''_{AB} = \frac{6EI}{L^2}\Delta = 2EK\left(\frac{3\Delta}{L}\right) \quad ; \quad M'''_{BA} = \frac{6EI}{L^2}\Delta = 2EK\left(\frac{3\Delta}{L}\right)$$

The final moments at the ends /joints can be obtained by taking the superimposed effect of the fixed end moment and different moment components due to rotation θ_A, θ_B and relative displacement Δ as follows

Thus:

$$M_{AB} = M_{AB}^{F} + M_{AB}^{'} + M_{AB}^{''} + M_{AB}^{'''}$$

$$M_{BA} = M_{BA}^{F} + M_{BA}^{'} + M_{BA}^{''} + M_{BA}^{'''}$$

In terms of the end rotations and relative displacement, the governing SD Equation as per the statical sign convention is formulated to be as follows:

$$M_{AB} = M_{AB}^{F} - 2EK\left(2\theta_A + \theta_B - 3\Delta/L\right)$$

$$M_{BA} = M_{BA}^{F} - 2EK\left(2\theta_B + \theta_A - 3\Delta/L\right)$$

SD Equation

Application of Slope-Deflection Equation

(i) Continuous beams with unyielding support:

For cases where there is no relative displacement between the two ends, the slope deflection equations can easily be modified by substituting $\Delta = 0$ in the general SD equation as follows:

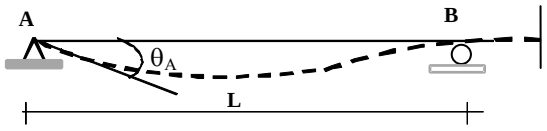
$$M_{AB} = M_{AB}^{F} - 2EK\left(2\theta_A + \theta_B\right)$$

$$M_{BA} = M_{BA}^{F} - 2EK\left(2\theta_B + \theta_A\right)$$

(ii) Continuous beams with yielding of supports

Yielding of support induces bending moments in continuous beam and the general slope-deflection equations formulated earlier are applicable for the case of specified rotations and displacements due to yielding of supports. Usually the effect of only yielding of support is investigated and the result may then be combined with those of applied loadings.

(iii) Beams with Hinged End



If one of the ends of the member is hinged, the slope deflection equation can be modified as follows.

$$M_{AB} = M^{F}_{AB} - 2EK\left(2\theta_A + \theta_B - 3\Delta/L\right) = 0$$

$$2EK\theta_A = \frac{M^{F}_{AB}}{2} - EK\theta_B + 3EK\frac{\Delta}{L} \qquad (*)$$

$$M_{BA} = M^{F}_{BA} - 2EK\left(2\theta_B + \theta_A - 3\Delta/L\right)$$

Substituting (*) in to the equation for M_{BA}

$$M_{BA} = M^{F}_{BA} - \frac{M^{F}_{AB}}{2} - 3EK\left(\theta_B - \Delta/L\right)$$

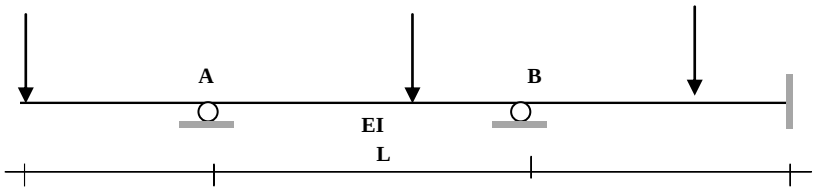
$$M_{BA} = M^{F1}_{BA} - 3EK\left(\theta_B - \Delta/L\right) \qquad ; M^{F1}{}_{BA} = M^{F}{}_{BA} - M^{F}{}_{AB}/2$$

Where: M^{F1}_{BA} is modified fixed end moment for the case of one end is fixed and the other is hinged.

The application of the above equation reduces the unknown θ values by one for a beam with hinge.

(iv) Beam with Overhanging End

Consider the following prismatic beam. Equilibrium condition gives:



Considering joint equilibrium condition at joint **A**: $M_{AB} + M_A = 0$

Where M_A is the cantilever moment due to the overhanging. The sign of M_A must be determined in accordance with the statical sign convention.

Thus

$$M_{AB} + M_A = M^F_{AB} - 2EK(2\theta_A + \theta_B - \frac{3\Delta}{L}) + M_A = 0$$

$$2EK\theta_A = \frac{1}{2}(M^F_{AB} + M_A) - EK\theta_B + \frac{3EK\Delta}{L}$$

Substituting into the second equation, the modified slope deflection equation for a beam with an overhang is:

$$M_{BA} = M^F_{BA} - \frac{1}{2}(M^F_{AB} + M_A) - 3EK(\theta_B - \frac{\Delta}{L})$$

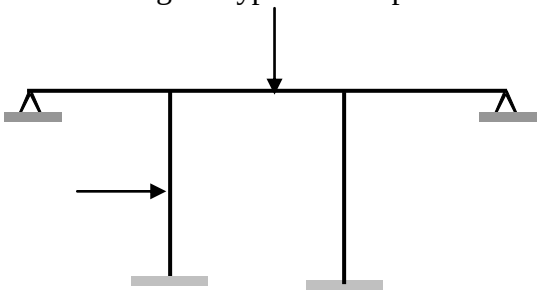
(v) Frames with out side-sway

A Frame may be considered to be non sway either if the frame is physically prevented from side sway by appropriate restraint in the direction of possibility of sway or if the frame is symmetrical and subjected to symmetric loading. In such cases the usual SD equation for members of the frame can be used with substation of zero for the term of lateral displacement component Δ and the SD equation reduces to

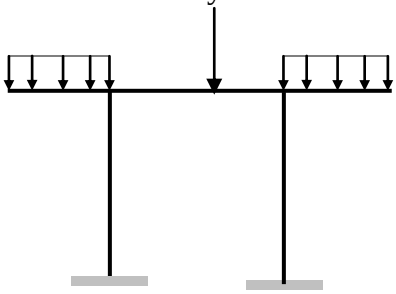
$$M_{AB} = M^F_{AB} - 2EK\left(2\theta_A + \theta_B\right)$$

$$M_{BA} = M^F_{BA} - 2EK\left(2\theta_B + \theta_A\right)$$

The following are typical examples for cases of frames with out side sway



Physically prevented from side sway



Symmetric frame subjected to symmetric loading

(Vi) Frame with Side Sway

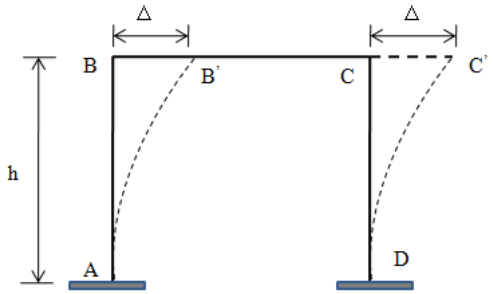
The usual SD equations are applicable and the following types of frames can be identified and the corresponding relative displacement of the ends can be considered in the SD equation. The governing slope deflection equations for members subjected to side sway are:

$$M_{AB} = M_{AB}^{F} - 2EK\left(2\theta_A + \theta_B - 3\Delta/L\right)$$

$$M_{BA} = M_{BA}^{F} - 2EK\left(2\theta_B + \theta_A - 3\Delta/L\right)$$

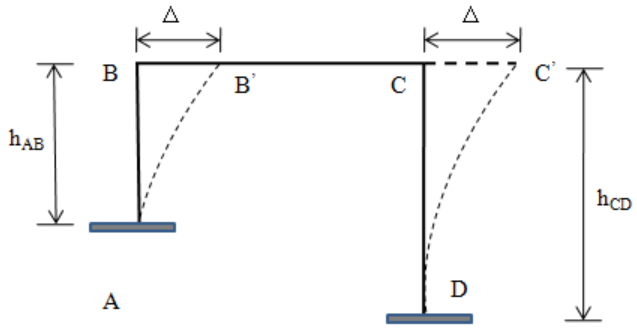
Assuming that the effect of axial deformations are small as compared to that of flexural strains, the relative side sway of the joints of members may be evaluated in terms of convenient lateral displacement depending on the geometry of the frame. Some cases of sway are now considered.

(a) Frame with Equal Heights.



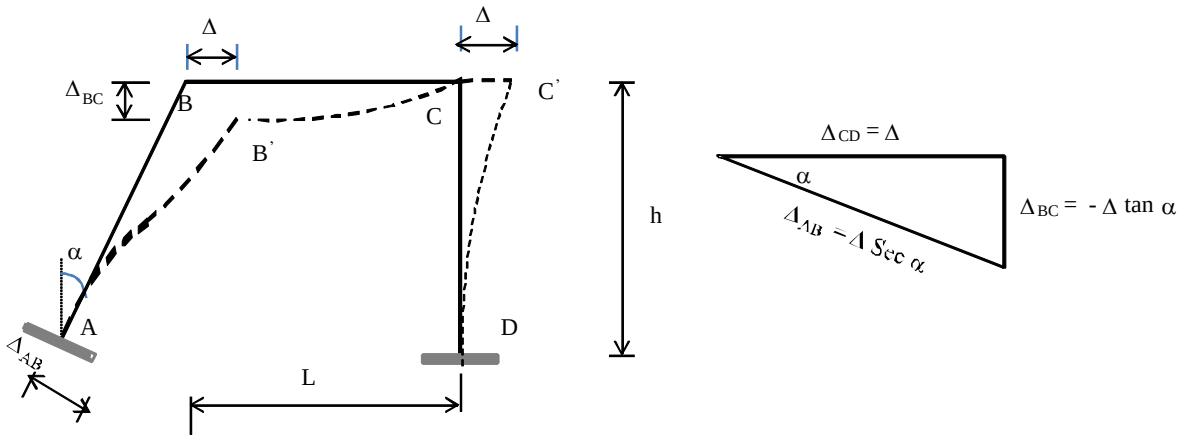
$$\left(\frac{\Delta}{L}\right)_{AB} = \left(\frac{\Delta}{L}\right)_{CD} = \frac{\Delta}{h} \;;\; \left(\frac{\Delta}{L}\right)_{BC} = 0$$

(b) Frame with unequal Heights.



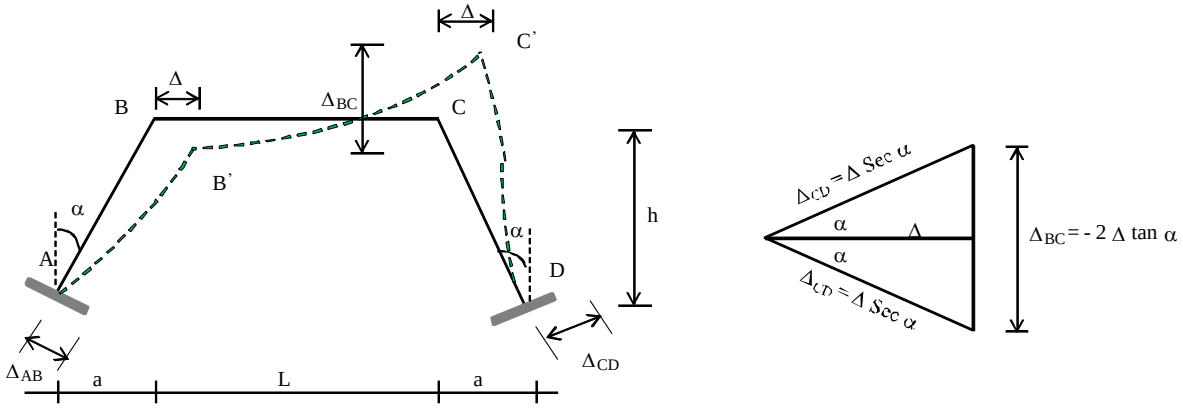
$$\left(\frac{\Delta}{L}\right)_{AB} = \frac{\Delta}{h_{AB}} \;,\; \left(\frac{\Delta}{L}\right)_{CD} = \frac{\Delta}{h_{CD}} \;,\; \left(\frac{\Delta}{L}\right)_{BC} = 0$$

(c) Frame with one inclined leg.



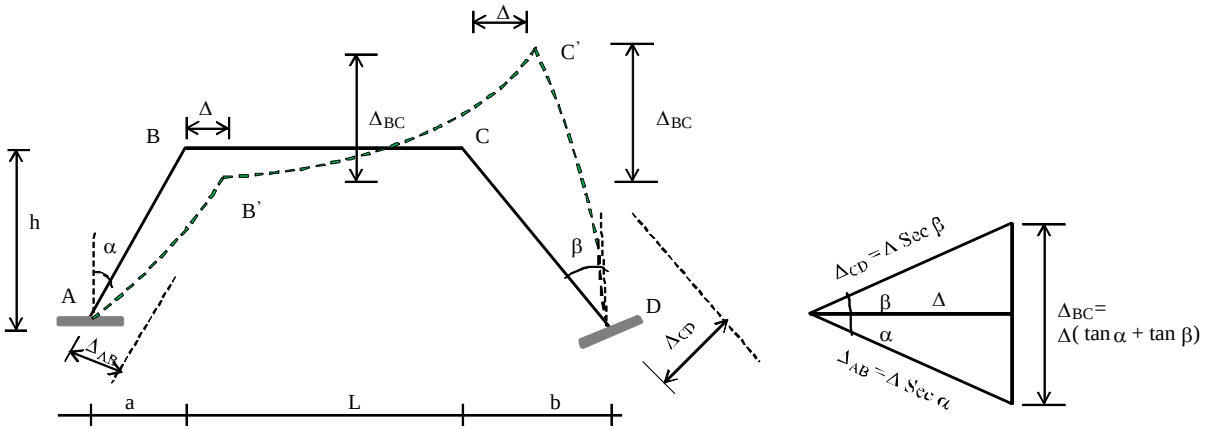
$$\left(\frac{\Delta}{L}\right)_{AB} = \frac{\Delta\, Sec\,\alpha}{h\, Sec\,\alpha} = \frac{\Delta}{h} \quad ; \quad \left(\frac{\Delta}{L}\right)_{BC} = -\frac{\Delta \tan\alpha}{L} \quad ; \quad \left(\frac{\Delta}{L}\right)_{CD} = \frac{\Delta}{h} \quad ,$$

(d) Symmetric Frame with inclined legs.



$$\left(\frac{\Delta}{L}\right)_{AB} = \left(\frac{\Delta}{L}\right)_{CD} = \frac{\Delta\, Sec\,\alpha}{h\, Sec\,\alpha} = \frac{\Delta}{h} \quad ; \quad \left(\frac{\Delta}{L}\right)_{BC} = -\frac{2\Delta \tan\alpha}{L}$$

(e) Unsymmetric Frame with inclined legs.



$\Delta_{AB} = \Delta \text{ Sec } \alpha$, $L_{AB} = h \text{ Sec } \alpha$ $\Delta_{BC} = \Delta (\tan \alpha + \tan \beta)$ thus;

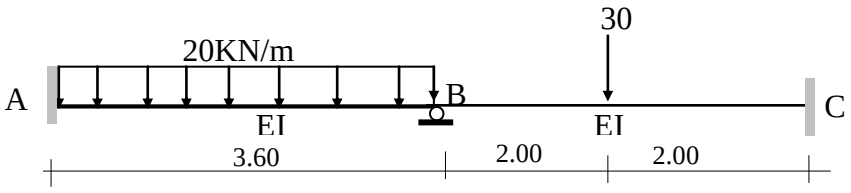
$$\left(\frac{\Delta}{L}\right)_{AB} = \frac{\Delta}{h} \quad ; \quad \left(\frac{\Delta}{L}\right)_{BC} = -\frac{\Delta (\tan\alpha + \tan\beta)}{L} = -\frac{a}{b} * \frac{\Delta}{h} \quad ; \left(\frac{\Delta}{L}\right)_{CD} = \frac{\Delta}{h}$$

Equilibrium Equations

Once the end moments of each member are expressed in terms of end rotations and relative displacement, sufficient number of equations need to be formulated by considering equilibrium condition of joints, members or the frame as a whole to solve the unknown rotations and displacements.

Example:

1. Determine the joint moments using the SD-equation for the following beam.



Soln.

$$M_{AB}^{F} = -M_{BA}^{F} = \frac{wL^2}{12} = \frac{20*3.6^2}{12} = 21.60$$

$$M_{BC}^{F} = -M_{CB}^{F} = \frac{PL}{8} = \frac{30*4}{8} = 15.00$$

$$\theta_A = \theta_C = \Delta_A = \Delta_B = \Delta_C = 0$$

$$M_{AB} = M_{AB}^{F} - 2EK\left(2\theta_A + \theta_B\right) = 21.6 - \frac{2EI}{3.6}\left(\theta_B\right) = 21.6 - 0.556EI\theta_B$$

$$M_{BA} = M_{BA}^{F} - 2EK\left(2\theta_B + \theta_A\right) = -21.6 - \frac{2EI}{3.6}\left(2\theta_B\right) = -21.6 - 1.111EI\theta_B$$

$$M_{BC} = M_{BC}^{F} - 2EK\left(2\theta_B + \theta_C\right) = 15.00 - \frac{2EI}{4.0}\left(2\theta_B\right) = 15.00 - 0.556EI\theta_B$$

$$M_{CB} = M_{CB}^{F} - 2EK\left(2\theta_C + \theta_B\right) = -15.00 - \frac{2EI}{4.0}\left(\theta_B\right) = -15.00 - 0.556EI\theta_B$$

Joint equilibrium condition: $\mathbf{M_{BA} + M_{BC} = 0}$

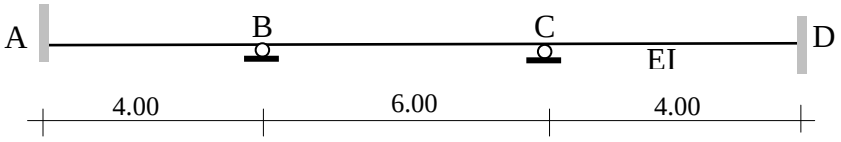
$$(-21.6 - 1.111EI\theta_B) + (15.00 - 0.556EI\theta_B) = 0 \rightarrow EI\theta_B = -3.126$$

Substituting the value of $EI\theta_B$ back in to the equation formulated above, the end moments are found to be:

$M_{AB} = +23.34$ $M_{BA} = -18.13$

$M_{BC} = +18.13$ $M_{CB} = -13.44$

2. Determine the support moments for the beam shown below due to the following support settlement: Support A rotates through 0.02 radian clockwise and support C settles down by 15mm. E = 210GPA, and I = $40*10^6$ mm^4.



Soln.

- Fixed end Moments: $M^F_{AB} = M^F_{BA} = M^F_{BC} = M^F_{CB} = M^F_{CD} = M^F_{DC} = 0$
- Slope deflection equations:

$$M_{AB} = M^F_{AB} - \frac{2EI}{4}\left(2\theta_A + \theta_B - \frac{3\Delta}{4}\right) = -\frac{2*8400}{4}\left(2*0.02 + \theta_B\right) = \underline{-168 - 4200\theta_B}$$

$$M_{BA} = M^F_{BA} - \frac{2EI}{4}\left(2\theta_B + \theta_A - \frac{3\Delta}{6}\right) = -\frac{2*8400}{4}\left(2\theta_B + 0.02\right) = \underline{-8400\theta_B - 84}$$

$$M_{BC} = M^F_{BC} - \frac{2EI}{4}\left(2\theta_B + \theta_C - \frac{3\Delta}{6}\right) = -\frac{2*8400}{6}\left(2\theta_B + \theta_C - \frac{3*0.015}{6}\right) = \underline{-5600\theta_B - 2800\theta_C + 21}$$

$$M_{CB} = M^F_{CB} - \frac{2EI}{6}\left(2\theta_C + \theta_B - \frac{3\Delta}{6}\right) = -\frac{2*8400}{6}\left(2\theta_C + \theta_B \frac{3*0.015}{6}\right) = \underline{-2800\theta_B - 5600\theta_C + 21}$$

$$M_{CD} = M^F_{CD} - \frac{2EI}{4}\left(2\theta_C + \theta_D - \frac{3\Delta}{4}\right) = -\frac{2*8400}{4}\left(2\theta_C - \frac{3*0.015}{4}\right) = \underline{-8400\theta_C - 47.25}$$

$$M_{DC} = M^F_{DC} - \frac{2EI}{4}\left(2\theta_D + \theta_C - \frac{3\Delta}{4}\right) = -\frac{2*8400}{4}\left(\theta_C - \frac{3*0.015}{4}\right) = \underline{-4200\theta_C - 47.25}$$

- Joint equation:

Joint B: $M_{BA} + M_{BC} = 0 \Rightarrow (-8400\theta_B - 84) + (-5600\theta_B - 2800\theta_C + 21) = 0$

$\Rightarrow -140000\theta_B - 2800\theta_C - 63 = 0$

$\Rightarrow -140000\theta_B - 2800\theta_C = 63$ **(1)**

Joint C: $M_{CB} + M_{CD} = 0 \Rightarrow (-2800\theta_B - 5600\theta_C + 21) + (-8400\theta_C - 47.25)$

$\Rightarrow -2800\theta_B - 14000\theta_C - 26.25 = 0$

$\Rightarrow -2800\theta_B - 14000\theta_C = 26.25$ **(2)**

- Solving equation **(1)** and **(2)** simultaneously;

θ_B = - 0.0043 rad. **θ_C = - 0.001 rad.**

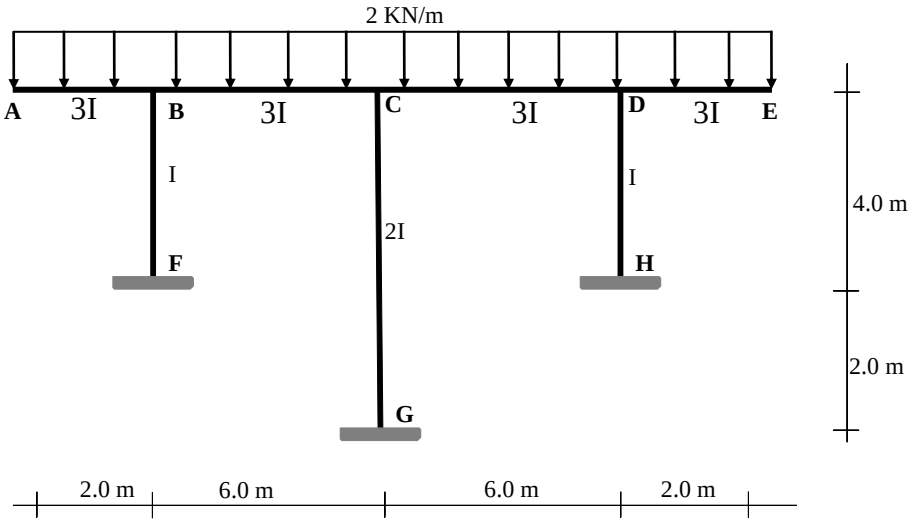
- Support moments:

$M_{AB} = -168 - 4200(-0043) = -150$ KN.m

$M_{BA} = -8400(-0.0043) - 84 = -47.9$ KN.m

M_{BC} = - 5600(-0.0043) – 2800(- 0.001) + 21 = + 47.9 KN.m
M_{CB} = - 2800(- 0.0043) – 5600(- 0.001) + 21 = + 38.7 KN.m
M_{CD} = - 8400(- 0.001) – 47.25 = - 38.8 KN.m
M_{DC} = - 4200(- 0.001) – 47.25 = + 43.1 KN.m

3. **U**sing the slope-deflection method, determine the support moments of the symmetric frame shown below



Solution:

- Relative stiffness:

Member	BC	CD	BF	CG	DH
I/L	3I/6	3I/6	I/4	2I/6	I/4
K	6	6	3	4	3

- Fixed-end moments

$$M_{BC}^{F} = -M_{CB}^{F} = M_{CD}^{F} = -M_{DC}^{F} = \frac{2*(6.0)^2}{12} = 6.0\ KN.m$$

- Slope-deflection Equations:

M_{FB} = - $6\theta_B$
M_{BF} = - $6(2\theta_B)$
M_{BC} = 6 – $12(2\theta_B + \theta_C)$, Where θ_C = 0(Symmetry)
M_{CB} = - 6 -$12\theta_B$
M_{CG} = - $8(2\theta_C + \theta_G)$ = 0, Where $\theta_C = \theta_G = 0$
M_{GC} = 0
M_{CD} = 6 – $12(2\theta_C + \theta_D)$, Where $\theta_C = 0$, $\theta_D = -\theta_B$
M_{DC} = - 6 – $12(2\theta_D)$, Where $\theta_D = -\theta_B$
M_{DC} = - 6 + $24\theta_B$
M_{DH} = - $6(2\theta_D)$ = $12\theta_B$
M_{HD} = - $6\theta_D$ = $6\theta_B$

- Joint equilibrium condition

i. Joint B: $M_{BF} + M_{BC} - 4 = 0$

ii. Joint C: $M_{CB} + M_{CD} + M_{CG} = 0$

iii. Joint D: $M_{DC} + M_{DH} + 4 = 0$

- Substituting the moment expressions into the equilibrium equation gives,

$- 12\theta_B + 6 - 24\theta_B - 4 = 0 \quad \Rightarrow \quad \mathbf{\theta_B = 0.0556} \quad ; \quad \mathbf{\theta_D = -0.0556}$

- End moments

$M_{FB} = - 6(0.0556) = - 0.333$ KN.m

$M_{BF} = - 6(2*0.0556) = - 0.667$ KN.m

$M_{BC} = 6 – 12(2*0.0556) = 4.667$ KN.m

$M_{CB} = - 6 – (12*0.0556) = - 6.667$ KN.m

$M_{CG} = M_{GC} = 0$

$M_{CD} = 6.667$ KN.m

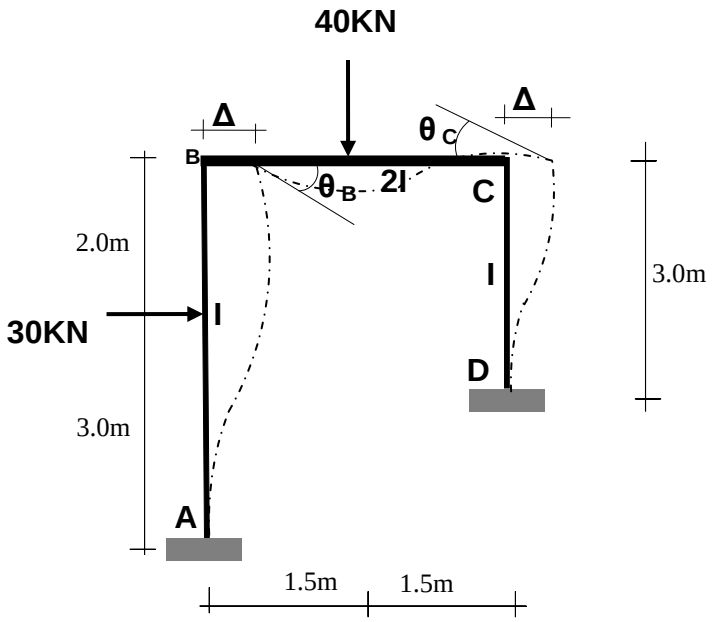
$M_{DC} = - 4.667$ KN.m

$M_{DH} = 0.667$ KN.m

$M_{HD} = 0.333$ KN.m

Example 4.

Determine the end moments using Sd equation and draw the shear force and bending moment diagram for the following frame



Relative Stiffness (Let I 15)

$K_{AB} = (I/L)_{AB} = 15/5 = 3.0$
$K_{BC} = (I/L)_{BC} = 2*15/3 = 10.0$
$K_{CD} = (I/L)_{CD} = 15/3 = 5.0$

Fixed End Moments

$$M^F_{AB} = \frac{30*3*2^2}{5^2} = +14.44$$

$$M^F_{BA} = \frac{30*2*3^2}{5^2} = -21.67$$

$$M^F_{BC} = -M^F_{CB} = \frac{40*3}{8} = +15.00$$

SD Equations:

$$M_{AB} = M^F_{AB} - 2EK\left(2\theta_A + \theta_B - 3\Delta/L\right) \quad ; \quad Let\ E = 1\ since\ all\ members\ are\ made\ of\ the\ same\ material$$

$$M^F_{AB} = +14.44 - 6\left(\theta_B - 0.6\Delta\right) \quad ; \qquad M^F_{BA} = -21.67 - 6\left(2\theta_B - 0.6\Delta\right)$$
$$M^F_{BC} = +15.00 - 20\left(2\theta_B + \theta_C\right) \quad ; \qquad M^F_{CB} = +15.00 - 20\left(\theta_B + 2\theta_C\right)$$
$$M^F_{CD} = 0.00 - 10\left(2\theta_C - \Delta\right) \quad ; \qquad M^F_{DC} = 0.00 - 10\left(\theta_C - \Delta\right)$$

(*)

Joint Equilibrium Conditions:

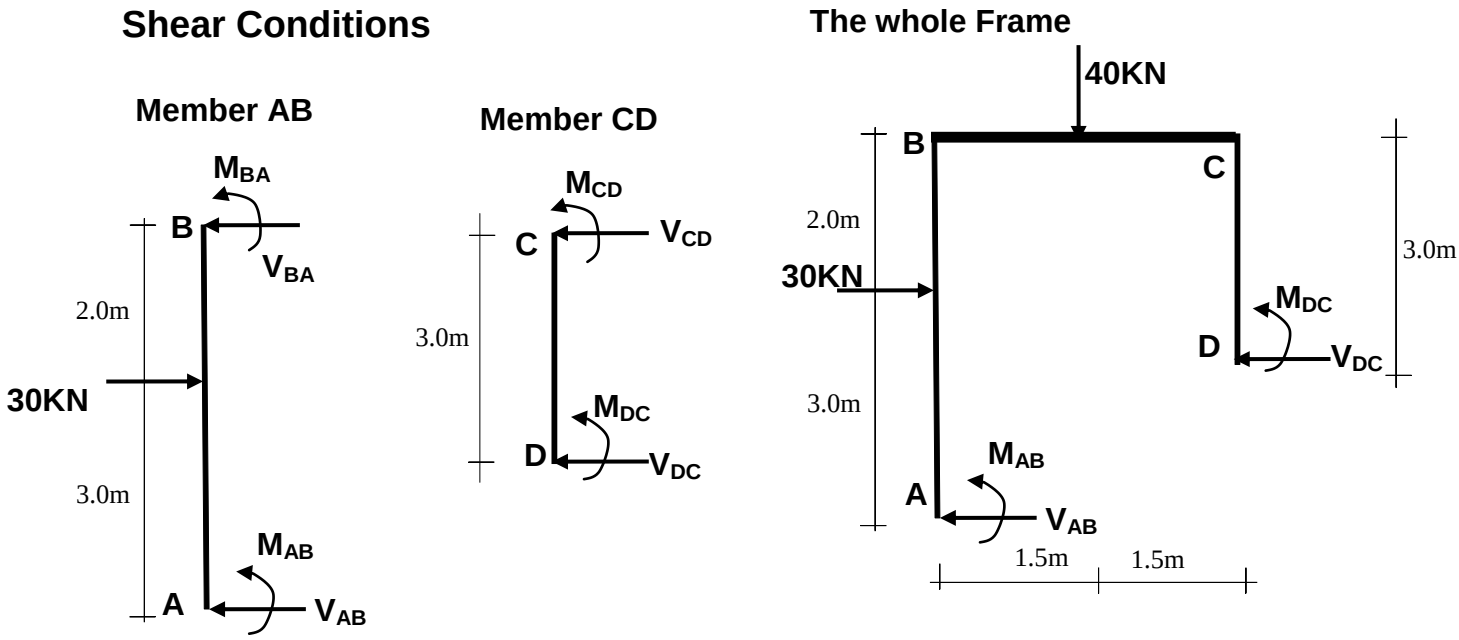
At Joint B: $M_{BA} + M_{BC} = 0$

$$-21.67 - 6\left(2\theta_B - 0.6\Delta\right) + 15.00 - 20\left(2\theta_B + \theta_C\right) = 0 \rightarrow 52\theta_B + 20\theta_C - 3.6\Delta = -6.67 \qquad (i)$$

At Joint C: $M_{CB} + M_{CD} = 0$

$$+15.00 - 20\left(\theta_B + 2\theta_C\right) + 0.00 - 10\left(2\theta_C - \Delta\right) = 0 \rightarrow 20\theta_B + 60\theta_C - 10\Delta = 15.00 \qquad (ii)$$

Shear Conditions



Member AB:

+) $\Sigma M_B = 0 : M_{AB} + M_{BA} + 30*2.0 - V_{AB} *5.0 = 0$

$$V_{AB} = \frac{M_{AB} + M_{BA} + 30*2.0}{5} = \frac{\left(14.44 - 6\left(\theta_B - 0.6\Delta\right)\right) + \left(-21.67 - 0.6\left(2\theta_B - 0.6\Delta\right)\right) + 60}{5} = -3.6\theta_B + 1.44\Delta + 10.554$$

Member CD:

+) $\Sigma M_C = 0 : M_{CD} + M_{DC} - V_{DC} *3.0 = 0$

$$V_{DC} = \frac{M_{CD} + M_{DC}}{3} = \frac{\left(0 - 10\left(2\theta_C - \Delta\right)\right) + \left(0 - 10\left(\theta_C - \Delta\right)\right)}{3} = -10\theta_C + 6.67\Delta$$

The whole Frame:

+ $\rightarrow$ $\Sigma F_H = 0 : 30 - V_{AB} - V_{DC} = 0$

$$30 - \left(-3.6\theta_B + 1.44\Delta + 10.554\right) - \left(-10\theta_C + 6.67\Delta\right) = 0$$

$$-3.6\theta_B - 10\theta_C + 8.11\Delta = 19.44 \qquad (iii)$$

Solving equations (i), (ii) and (iii) simultaneously:

$\theta_B = -0.002$; $\theta_B = 0.020$; $\Delta = 0.263$

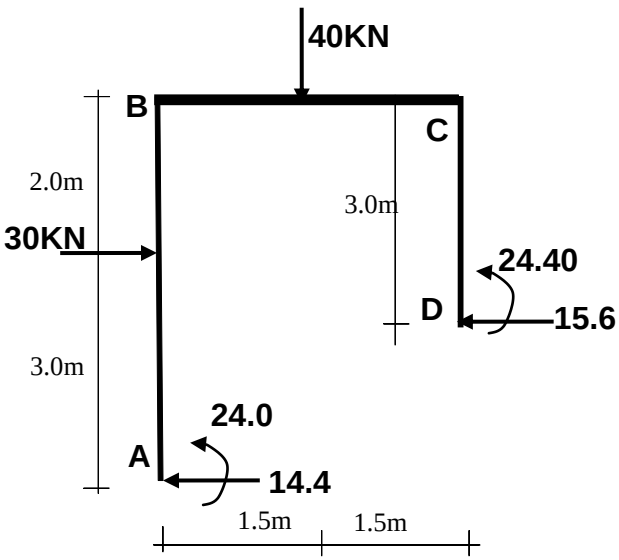
Substituting the values of θ_B , θ_B and Δ in to (*)

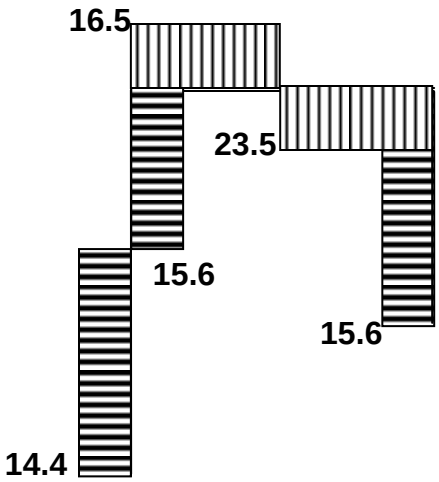
$M_{AB} = 24.0KNm$ $M_{BA} = -11.90KNm$
$M_{BC} = 11.90KNm$ $M_{CB} = -22.40KNm$
$M_{CD} = 22.40Nm$ $M_{Dc} = 24.40KNm$

Substituting the values of θ_B , θ_C and Δ in to the repective equation for V_{AB} and V_{DC}

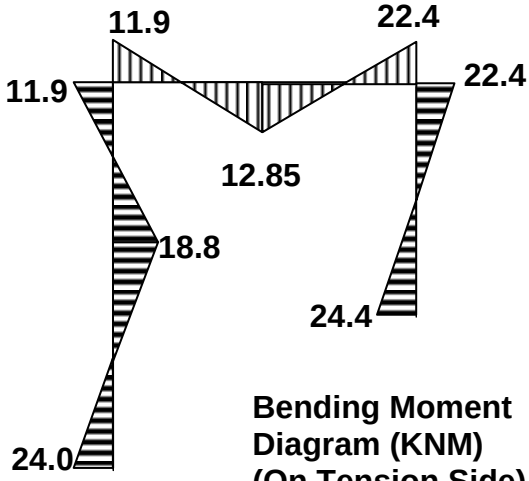
$V_{AB} = 14.4KN$ $V_{DC} = 15.6KN$

Shear force and Bending Moment Diagrams





Shear Force Diagram (KN)



Bending Moment Diagram (KNM) (On Tension Side)

Eexercise

Determine the support moments using SD equation and draw the shear force and bending moment diagrams for the following frames.

